

Hornsby Girls' High

Mathematics Preliminary Examination – 2004

QUESTION 1 (11 marks) (START A NEW PAGE) Marks

(a) Factorise completely

(i) $x^2 - 9$ 1

(ii) $27 - 81^3$ 1

(b) Evaluate correct to 2 significant figures

$$\sqrt[4]{\frac{86.54 \times 3.16}{2.6^3}}$$
 2

(c) Given that triangle ABC is a rightangled triangle in which $\angle ABC = 90^\circ$, $AB = (2\sqrt{3} - 1)$ cm and $BC = (\sqrt{3} + 2)$ cm. Find the length of AC, leaving your answer in simplest surd form. 3

(d) Convert $0.\dot{2}1\dot{2}$ into a fraction in its simplest form. 2

(e) At a nursery, all plants increased in price by 35%. One plant now costs \$82.50. What was the old price for this plant? 2

QUESTION 2 (9 marks) (START A NEW PAGE)

(a) Solve for x : $\frac{2x+3}{3} - \frac{3x-1}{4} = -3$ 2

(b) Simplify: $\frac{x^2 - 2x}{y^2 - 4y + 4} \div \frac{x}{y-2}$ 3

(c) The quadratic equation $3x^2 - 2x - 5 = 0$ has roots α and β . Without finding the roots evaluate:

(i) $\alpha\beta^2 + \beta\alpha^2$ 2

(ii) $(\alpha - 3)(\beta - 3)$ 2

QUESTION 3 (10 marks) (START A NEW PAGE)

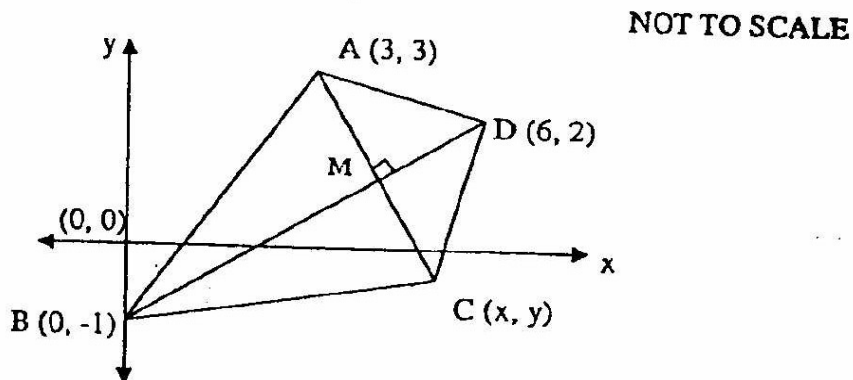
Marks

- (a) For the parabola $x^2 + 6x - 12y + 3 = 0$
- (i) Find the coordinates of the vertex. 2
 - (ii) Calculate the focal length. 1
 - (iii) Find the coordinates of the focus. 1
 - (iv) Find the equation of the directrix 1
- (b) Find the value(s) of k for which $x^2 + (k-1)x + 4 = 0$ has real roots. 2
- (c) Given $\frac{8}{x^2 + 5x + 4} \equiv \frac{a}{(x+4)} + \frac{bx+c}{(x+1)}$ find the values of a , b and c . 3

QUESTION 4 (10 marks) (START A NEW PAGE)

Marks

- (a) In the quadrilateral ABCD, the points A, B and D are (3, 3), (0, -1) and (6, 2) respectively. The line BD bisects the line AC at right angles at the point M.



- (i) Find the distance BD. 1
- (ii) Show that the gradient of BD is $\frac{1}{2}$. 1
- (iii) Show that the equation of line BD is $x - 2y - 2 = 0$. 1

Question 4 (CONT)**Marks**

(iv) Show that the equation of AC is $2x + y - 9 = 0$. 1

(v) Find the coordinates of M. 2

(vi) Hence, find the coordinates of C 1

(b) A function $f(x)$ is defined as:
$$f(x) = \begin{cases} 2 - x & \text{for } x < -3 \\ 5 & \text{for } -3 \leq x < 0 \\ x^2 - 1 & \text{for } x \geq 0 \end{cases}$$

Calculate the value of (a) $f(-10)$ 1

(b) $f(-3)$ 1

(c) $f(a^2)$ 1

QUESTION 5 (11 marks) (START A NEW PAGE)

(a) (i) Sketch $f(x) = 4 + \frac{1}{1-x}$ showing any asymptotes and intercepts with the axes. 2

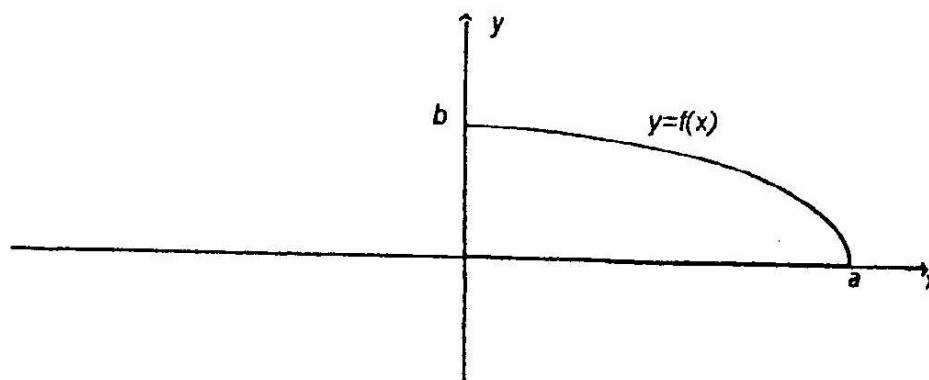
(ii) State the range of the above function. 1

(b) Shade the region represented by

$$4x + 3y - 12 < 0 \text{ and } y \geq 3^x + 1$$

(Note: Do not find the point of intersection of the two curves)

3

Question 5 (continued)**Marks****(c)**

The diagram above shows part of the graph of the function $y = f(x)$. You are told that it is an odd function.

1

Copy the diagram and complete the graph of the function

- (d) The point $P(x, y)$ moves so that its distance PA from the point $A(1, 5)$ is always twice its distance PB from the point $B(4, -1)$.

(i) Show that the locus is a circle with equation $x^2 + y^2 - 10x + 6y + 14 = 0$

2

(ii) Find the centre and radius of this circle

2**QUESTION 6 (7 marks) (START A NEW PAGE)**

- (a) If $\cos \alpha = \frac{2}{7}$ and $\sin \alpha < 0$, find the exact value of $\tan \alpha$ in the domain $0 \leq \alpha \leq 360^\circ$

2

- (b) Angela finds the angle of elevation of the top of a building to be 35° . After walking on horizontal ground for 60m towards the building she now finds the angle of elevation to be 70° .

(i) Draw a neat sketch showing this information.

1

(ii) Find the height of the building to the nearest metre.

2

- (c) Evaluate $\cos 135^\circ + \operatorname{cosec} 60^\circ$, in exact form.

2

QUESTION 7 (10 marks) (START A NEW PAGE)

Marks

- (a) Two surveyors, Phil and Chris place their surveying equipment at point X then set out in different directions. Phil walks due West for 200 metres while Chris walks 150 metres on a bearing of 130° .

- (i) Draw a diagram showing the above information. 1
- (ii) Calculate the distance between Phil and Chris (to the nearest metre) 2
- (iii) Find the bearing of point X from Chris. 1
- (iv) Calculate the bearing of Phil from Chris (to the nearest degree) 2

- (b) (i) Show that: $\tan \theta + \cot \theta = \frac{1}{\sin \theta \cos \theta}$ 1
- (ii) Hence or otherwise, solve: 3

$$\frac{1 + \cot \theta}{\operatorname{cosec} \theta} - \frac{\sec \theta}{\tan \theta + \cot \theta} = -1, \quad 0^\circ \leq \theta \leq 360^\circ$$

QUESTION 8 (9 marks) (START A NEW PAGE)

- (a) Find the derivatives of:

- (i) $\sqrt{x^3} + \frac{1}{2x}$ 2
- (ii) $(3x^2 - 4x)^3$ 2
- (iii) $\frac{x+3}{x^2-4}$ 2

- (b) A tangent is drawn to the parabola $y = x^2 - 4x$ at the point P. The tangent has a gradient of 6.

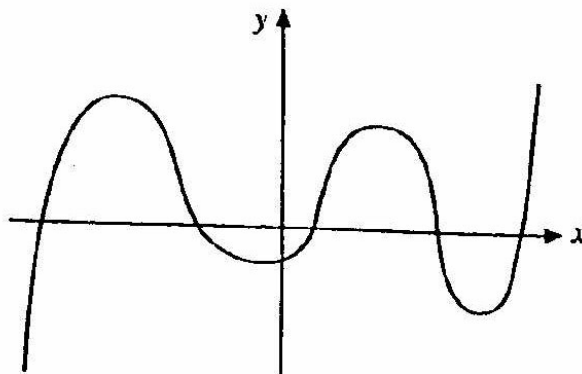
- (i) Find the coordinates of P. 2
- (ii) Find the equation of the tangent at P. 1

QUESTION 9 (16 marks) (START A NEW PAGE)

Marks

- (a) Find the values of x for which the curve $y = 2x^3 - 5x^2$ is decreasing **1**

- (b) Copy the graph below and sketch the derivative function on the same graph **1**



- (c) Differentiate $f(x) = 2x^2 - 7x$ from first principles **3**

- (d) Sketch the graph of $y = x(2 - x)^3$ by finding the stationary points and determining their nature, the points of inflection and x-intercepts. On your sketch clearly show all important features. **5**

- (e) (i) The sum of the radii of two circles is 100 cm. If one of the circles has a radius of x cm, show that the sum of the areas of the two circles is given by $A = 2\pi(x^2 - 100x + 5000)$ **2**

- (ii) Find the value of x for which A is the least. **3**

- (f) Find the primitive function of x^{-3} **1**

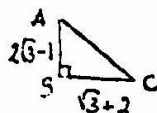
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HGHS 2004 20 Yearly

1a) i) $(x-3)(x+3)$

ii) $3^3 - (2y)^3 = (3-2y)(9+6y+4y^2)$

b) $1.9860... = 2.0$

c) 

$$AC^2 = (2\sqrt{3}-1)^2 + (\sqrt{3}+2)^2$$

$$= 4 \times 3 - 4\sqrt{3} + 1 + 3 + 4\sqrt{3} + 4$$

$$= 20$$

$$\therefore AC = \sqrt{20} \quad (\sqrt{4 \times 5})$$

$$= 2\sqrt{5}$$

d) let $x = 0.212212212...$

$1000x = 212.212212...$

$\therefore 999x = 212$

$x = 212/999$

e) 135% is \$82.50

1% is $82.50/135$

$\therefore 100\%$ is $82.50/135 \times 100$

$= 61.111...$

$\therefore$ old price was \$61.11

2a) $12x \frac{(2x+3)}{3} - 12 \frac{(3x-1)}{4} = 12x-3$

$8x+12-9x+3 = -36$

$-x+15 = -36$

$-x = -51$

$\therefore x = 51$

b) $\frac{x(x-2)}{(y-2)(y-2)} \times \frac{y-2}{x}$

$= \frac{x-2}{y-2}$

c) $\alpha + \beta = -\frac{b}{a}$ $\alpha\beta = \frac{c}{a}$

$= \frac{2}{3}$

$= -\frac{5}{3}$

i) $\alpha\beta(\beta+\alpha) = -\frac{5}{3} \times \frac{2}{3}$

$= -\frac{10}{9}$

ii) $\alpha\beta - 3\alpha - 3\beta + 9 = \alpha\beta - 3(\alpha+\beta) + 9$

$= -\frac{5}{3} - 3 \times \frac{2}{3} + 9$

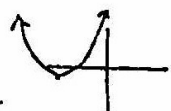
$= 5\frac{1}{3}$

3a) $x^2 + 6x + \left(\frac{6}{2}\right)^2 = 12y - 3 + 9$

$(x+3)^2 = 12y + 6$

$(x+3)^2 = 12\left(y + \frac{1}{2}\right)$

i) vertex: $(-3, -\frac{1}{2})$



ii) $4a = 12 \therefore a = 3$

iii) focus: $(-3, 2\frac{1}{2})$

iv) directrix: $y = -3\frac{1}{2}$

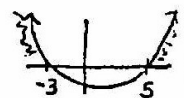
b) real roots $\Delta \geq 0$

$\Delta = (k-1)^2 - 4 \times 1 \times 4$

$= k^2 - 2k + 1 - 16$

$= k^2 - 2k - 15$ $\frac{k-5}{k+3}$

$= (k-5)(k+3)$



so $(k-5)(k+3) \geq 0$

$\therefore k \geq 5, k \leq -3$

c) $\frac{8}{(x+4)(x+1)} = \frac{a(x+1) + (bx+c)(x+4)}{(x+4)(x+1)}$

$\therefore 8 \equiv ax+a+bx^2+4bx+cx+4c$

$\equiv bx^2 + (a+4b+c)x + a+4c$

$\therefore b=0, a+4b+c=0$, so $a+4c=8$

$\therefore a+c=0$ $-c+4c=8$

$a=-c$ $\therefore 3c=8$

$\therefore c=8/3$

$\therefore a = -8/3 (-2\frac{2}{3}), b=0, c=8/3 (2\frac{2}{3})$

4a) i) $BD = \sqrt{(c-0)^2 + (2-1)^2}$

$= \sqrt{36+9}$

$= \sqrt{45}$

$= 3\sqrt{5}$

$$\text{ii) } m_{BD} = \frac{2-1}{6-0} \\ = \frac{1}{6}$$

$$\text{iii) } y-1 = \frac{1}{2}(x-0) \\ y+1 = \frac{1}{2}x \\ 2y+2 = x \\ \therefore 0 = x-2y-2$$

$$\text{iv) } m_{AC} = -2$$

$$\therefore y-3 = -2(x-3) \\ y-3 = -2x+6 \\ 2x+y-9 = 0$$

$$\text{v) } \begin{aligned} x-2y-2 &= 0 \quad \dots \textcircled{1} \\ 2x+y-9 &= 0 \quad \dots \textcircled{2} \\ \textcircled{1} \times 2 \quad 2x-4y-4 &= 0 \quad \dots \textcircled{3} \\ \textcircled{2}-\textcircled{3} \quad 5y-5 &= 0 \\ 5y &= 5 \\ y &= 1 \end{aligned}$$

$$\text{sub in } \textcircled{1} \quad x-2-2=0 \\ \therefore x=4$$

So M is (4,1)

$$\text{vi) } \text{H is midpoint} \\ \therefore 4 = \frac{x+3}{2}, \quad 1 = \frac{y+3}{2} \\ 8 = x+3 \quad 2 = y+3 \\ \therefore x=5 \quad \therefore y=-1$$

so C has coords (5,-1)

$$\text{b) (a) } f(-10) = \frac{2-10}{6-0} \\ = \frac{-8}{6}$$

$$\text{(b) } f(-3) = \frac{2-1}{6-0} = \frac{1}{6}$$

$$\text{(c) } f(a^2) = \frac{2-a^2}{6-0} = \frac{2-a^2}{6}$$

$$\text{5a) } f(x) = 4 + \frac{1}{1-x}$$

$$\text{asymptotes: } x=1$$

$$y-4 = \frac{1}{1-x}$$

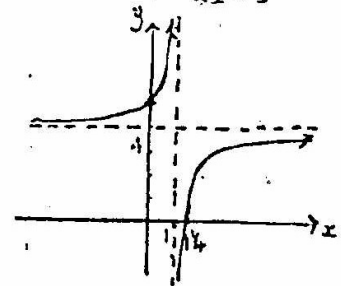
$$1-x = \frac{1}{y-4}$$

$$\therefore x = 1 - \frac{1}{y-4}$$

$$\therefore \text{asymptote: } y=4$$

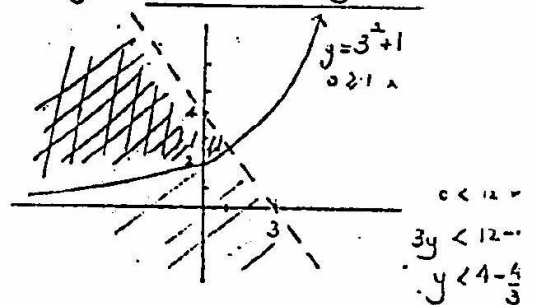
$$y\text{-int: } (1=0) \\ \therefore y=5$$

$$x\text{-int: } (y=0) \quad 0 = 4 + \frac{1}{1-x} \\ 0 = 4(1-x) + 1 \\ 0 = 4 - 4x + 1 \\ 4x = 5$$

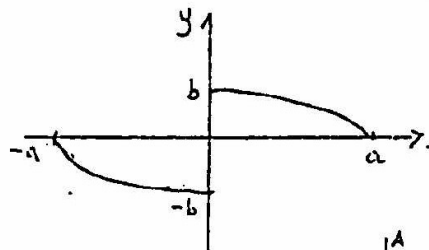


ii) range: all real $y, y \neq 4$

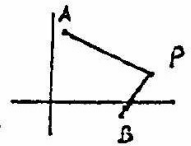
b)



c)



d) i) $PA = 2PB$

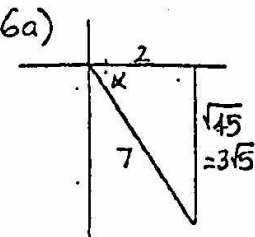


$$\begin{aligned} \sqrt{(x-1)^2 + (y-5)^2} &= 2\sqrt{(x-4)^2 + (y+1)^2} \\ x^2 - 2x + 1 + y^2 - 10y + 25 &= 4[x^2 - 8x + 16 + y^2 + 2y + 1] \\ x^2 - 2x + y^2 - 10y + 26 &= 4x^2 - 32x + 64 + 4y^2 + 8y + 4 \\ 0 &= 3x^2 - 30x + 3y^2 + 18y + 42 \\ \therefore 0 &= x^2 - 10x + y^2 + 6y + 14 \end{aligned}$$

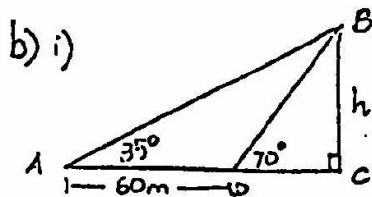
$$\text{ii) } -14 + 25 + 9 = x^2 - 10x + \left(\frac{-10}{2}\right)^2 + y^2 + 6y + \left(\frac{6}{2}\right)^2 \\ 20 = (x-5)^2 + (y+3)^2$$

$\therefore$ circle centre $(5, -3)$, radius $= \frac{\sqrt{20}}{2} = \frac{\sqrt{5}}{2}$

6a)



$$\therefore \tan \alpha = -\frac{3\sqrt{5}}{2}$$



ii) $\angle ABD = 35^\circ$ (ext $\angle$ of Δ -theorem)
 $\therefore \Delta ABD$ is isosceles
 $\therefore BD = 60 \text{ m}$

$$\therefore \sin 70^\circ = \frac{h}{60}$$

$$\therefore h = 60 \sin 70^\circ$$

$$= 56.38155 \dots$$

$$= 56 \text{ m}$$

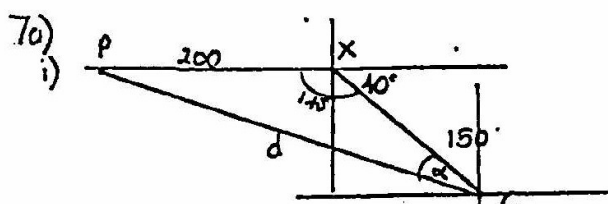
c) $\cos 135 = -\cos(180-45)$, $\operatorname{cosec} 60 = \frac{1}{\sin 60}$

$$= -\cos 45 = \frac{1}{\frac{\sqrt{3}}{2}} = \frac{2}{\sqrt{3}}$$

$$= -\frac{1}{\sqrt{2}}$$

$$\therefore \cos 135 + \operatorname{cosec} 60 = \frac{-1}{\sqrt{2}} + \frac{2}{\sqrt{3}}$$

$$= \frac{-\sqrt{3} + 2\sqrt{2}}{\sqrt{6}}$$



ii) $d^2 = 200^2 + 150^2 - 2 \times 200 \times 150 \times \cos 140$

$$= 108462.66 \dots$$

$$\therefore d = \sqrt{108462.66 \dots}$$

$$= 329.336 \dots$$

$$\approx 329 \text{ m}$$

iii) $360 - 50 = 310^\circ$ (or $270 + 40$)

iv) $\frac{\sin \alpha}{200} = \frac{\sin 140}{329}$

$$\therefore \sin \alpha = \frac{200 \sin 140}{329}$$

$$\sin \alpha = 0.3907 \dots$$

$$\therefore \alpha = 23$$

$\therefore$ bearing of Phil from Chris :

$$310 - 23 = 287^\circ$$

b) i) LHS = $\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}$

$$= \frac{\sin^2 \theta + \cos^2 \theta}{\cos \theta \sin \theta}$$

$$= \frac{1}{\cos \theta \sin \theta}$$

$$= \text{RHS}$$

ii) LHS: $1 + \frac{\cos \theta}{\sin \theta} - \frac{1}{\cos \theta}$

$$= \frac{\sin \theta + \cos \theta}{\sin \theta} - \frac{1}{\cos \theta}$$

$$= \left(\frac{\sin \theta + \cos \theta}{\sin \theta} \times \frac{\sin \theta}{1} \right) - \left(\frac{1}{\cos \theta} \times \frac{\sin \theta \operatorname{cosec} \theta}{1} \right)$$

$$= \sin \theta + \cos \theta - \sin \theta$$

$$= \cos \theta$$

$$\therefore \cos \theta = -1$$

$$\therefore \theta = 180^\circ$$



8a) i) let $y = x^{3/2} + \frac{1}{2} x^{-1}$

$$y' = \frac{3}{2} x^{1/2} - \frac{1}{2} x^{-2}$$

$$= \frac{3\sqrt{x}}{2} - \frac{1}{2x^2}$$

ii) $\frac{5(3x^2-4x)^4(6x-4)}{10(3x^2-4x)^4(3x-2)}$

iii) let $y = \frac{x+3}{x^2-4}$, $u = x+3$, $v = x^2-4$

$$u' = 1 \quad v' =$$

$$y' = \frac{(x^2-4) \times 1 - (x+3)(2x)}{(x^2-4)^2}$$

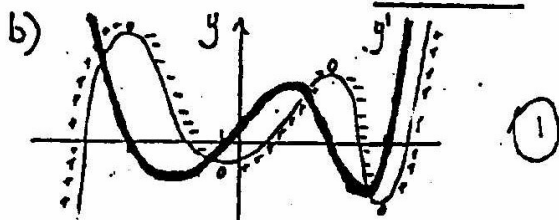
$$= \frac{x^2-4-2x^2-6x}{(x^2-4)^2}$$

$$= \frac{-x^2-6x-4}{(x^2-4)^2}$$

$\therefore m = 6, y' = 2x - 4$
 $\therefore 2x - 4 = 6$
 $2x = 10$
 $x = 5 \quad \therefore y = 5^2 - 4 \times 5$
 $\therefore P \text{ has co-ords } (5, 5)$

ii) $y - 5 = 6(x - 5)$
 $y = 6x - 25 \text{ or } 6x - y - 25 = 0$

9a) decreasing $y' < 0$,
 $y' = 6x^2 - 10x$
 $= 2x(3x - 5)$
 $\therefore 2x(3x - 5) < 0$
 $\text{ie. } 0 < x < \frac{5}{3}$ (1)



b) $f(x) = 2x^2 - 7x, f(x+h) = 2(x+h)^2 - 7(x+h)$
 $= 2(x^2 + 2xh + h^2) - 7x - 7h$
 $= 2x^2 + 4xh + 2h^2 - 7x - 7h$

$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$
 $= \lim_{h \rightarrow 0} \frac{2x^2 + 4xh + 2h^2 - 7x - 7h - (2x^2 - 7x)}{h}$
 $= \lim_{h \rightarrow 0} \frac{4xh + 2h^2 - 7h}{h}$
 $= 4x - 7$ (3)

d) $y = x(2-x)^3$
 $y' = x \times 3(2-x)^2 \times (-1) + (2-x)^3$
 $= -3x(2-x)^2 + (2-x)^3$
 $= (2-x)^2(-3x + 2 - x)$
 $= (2-x)^2(2 - 4x)$
 $= 2(2-x)^2(1 - 2x)$ (1)

stat pts: $y' = 0$ ie. $x = 2, x = \frac{1}{2}$

$y'' = 2(2-x)^2 \times (-1) + 2 \times 2(2-x) \times (-1) \times (1-2x)$
 $= -4(2-x)^2 - 4(2-x)(1-2x)$
 $= -4(2-x)(2-x + 1-2x)$
 $= -4(2-x)(3-3x)$
 $= -12(2-x)(1-x)$

when $x = 2, y = 0$
 $y'' = 0$
 $\therefore$ poss. pt of inf.

when $x = \frac{1}{2}, y = \frac{27}{16}$
 $y'' = -9 < 0$
 $\therefore (\frac{1}{2}, \frac{27}{16})$ max. p.

now. poss. pts of inflection $y'' = 0$

ie. $-12(2-x)(1-x) = 0$

$\therefore x = 2, x = 1$

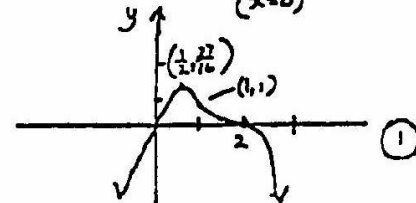
x	2	1	2
y'	+	0	-

$\therefore (2, 0)$ is a horizontal pt. of inf.

x	1	1	1
y''	-	0	+

$\therefore (1, 1)$ is a pt of inf.

x -int: $x = 0, x = 2$
 $(y = 0)$



e) i) $A_1 = \pi x^2$

$A_2 = \pi(100 - x)^2$

$\therefore A = \pi x^2 + \pi(10000 - 200x + x^2)$
 $= \pi(x^2 + 10000 - 200x + x^2)$
 $= \pi(2x^2 + 10000 - 200x)$
 $= 2\pi(x^2 - 100x + 5000)$ (2)

ii) $A' = 2\pi(2x - 100)$
 $= 4\pi(x - 50)$ (1)

max when $A' = 0$
 ie. $4\pi(x - 50) = 0$
 when $x = 50$ (1)

now $A'' = 4\pi > 0$ (1)

$\therefore x = 50 \text{ cm}$ is least value.

f) $\frac{x^{-2}}{-2} + c$ or $-\frac{1}{2x^2} + c$ (1)